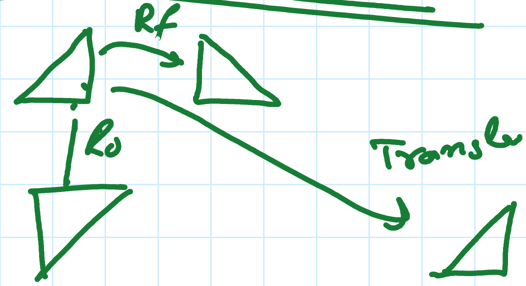
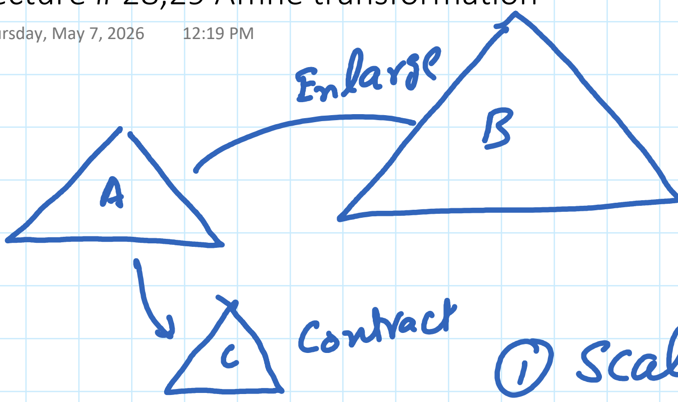
Lecture # 28,29 Affine transformation

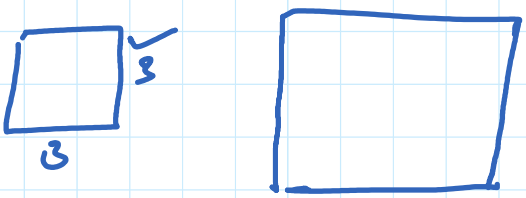
Thursday, May 7, 2026 12:19 PM

Eucldean Transforms

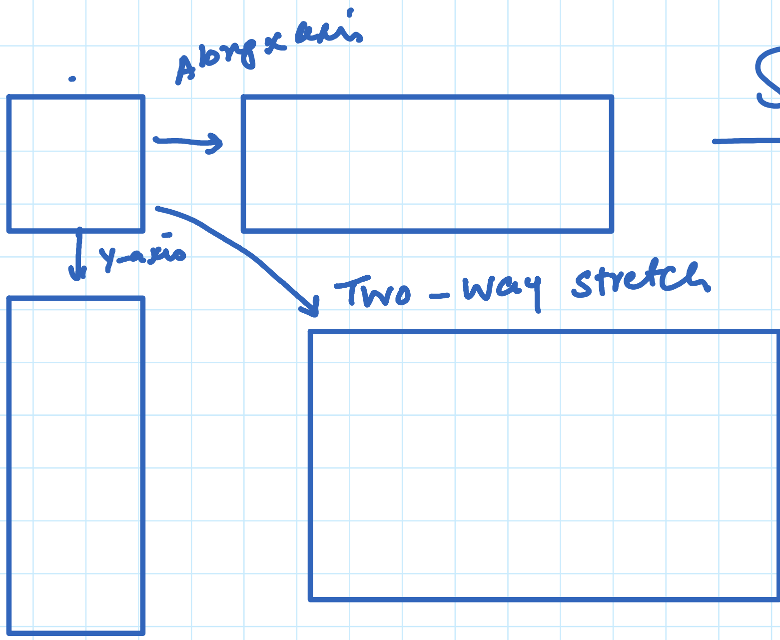




(i) Scaling ⇒ Similar shape



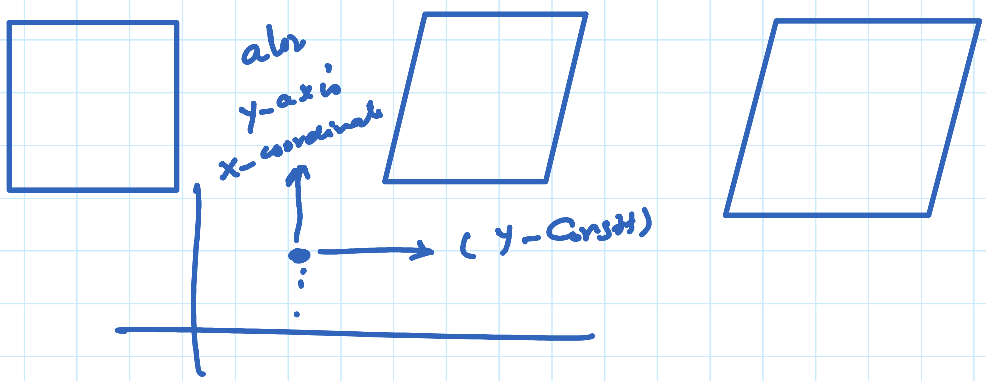
(ii) Stretching



(iii) SHEAR (H)

Along x-axis

Along y-axis



How to do

① **Scaling** $(0,0)$; $K \Rightarrow$ Scale factor $+$, $-$

$-1 < K < 1$ Contraction

$-1 > K > 1$ Enlargement

* Object, Image, C.O.F are Colinear

Scale factor $= \dfrac{\text{Image length}}{\text{Object length}}$

$\dfrac{IC}{OC}$

+ve { * O * I * C d_1 d_2

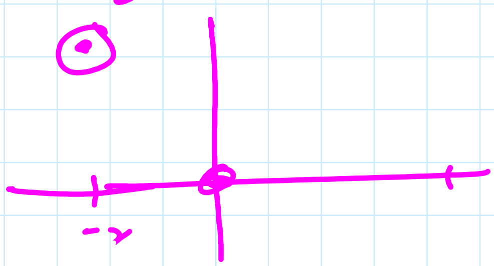
* I * O * C d_o d_i

$d_i > d_o$

−ve (I C O

$(x, y) \xrightarrow{K} (Kx, Ky)$

$(-2, 3) \xrightarrow{2} (-4, 6)$,



$(-2, 3) \xrightarrow{-2} (4, -6)$

Matrix $\begin{pmatrix} K & 0 \\ 0 & K \end{pmatrix}$

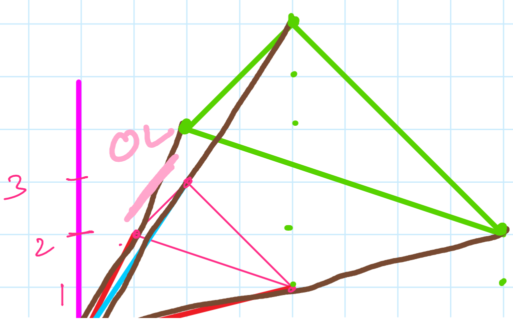
Co-ordinates function

$(x, y) \xrightarrow{K} (Kx, Ky)$

A(1, 2) A′(−2, −4)

B(4, 1) B′(−8, −2)

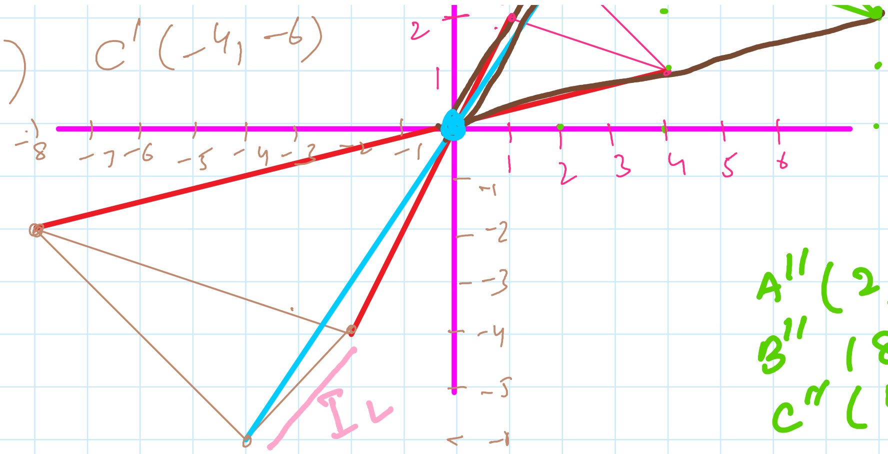
C(2, 3) C′(−4, −6)



Scale factor = −2

$K = +2$

C (2,3) C' (-4, -6)



$-\frac{IL}{OL}$

A'' (2, 4)
B'' (8, 2)
C'' (4, 6)

$$\frac{(x-2)^2}{4} + \frac{(y-1)^2}{9} = 1$$

Ellipse

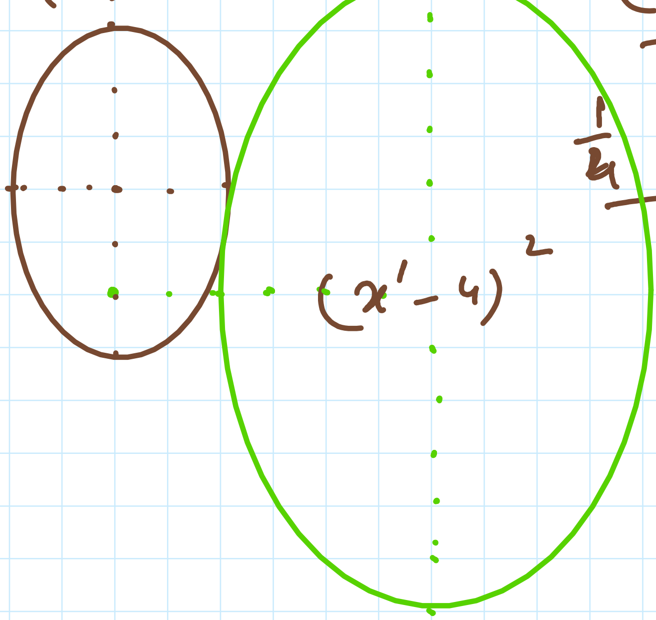
scal. 2

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$x' = 2x \longrightarrow x = \frac{1}{2}x'$

$y' = 2y \longrightarrow y = \frac{1}{2}y'$

(2, 1)



$$\frac{(\frac{1}{2}x'-2)^2}{4} + \frac{(\frac{1}{2}y'-1)^2}{9} = 1$$

$$\frac{\frac{1}{4}(x'-4)^2}{4} + \frac{\frac{1}{4}(y'-2)^2}{9} = 1$$

$$\frac{(x'-4)^2}{4^2} + \frac{(y'-2)^2}{6^2} = 1$$